

From Singapore to Rotterdam, digital twins are helping cities plan the future. In Dharavi, surveyors are mapping one of the world's most complex urban landscapes.

A Virtual Dharavi is Taking Shape Before a New City Rises

Before new roads, homes and public spaces emerge, a detailed digital replica is helping planners navigate one of India's most ambitious urban transformations

From Singapore and Heathrow Airport in the United Kingdom to Rotterdam in the Netherlands and Helsinki in Finland, digital twins have emerged as one of the world's most powerful urban-planning tools. Governments use them to model climate risks, airports rely on them to improve operations and city administrators deploy them to test infrastructure decisions before construction begins.

A Different Urban Challenge

Long before new buildings emerge and wider roads take shape, a detailed virtual replica of one of the world's most densely populated informal settlements is being created. Spread across 621 acres and home to nearly one million people, the settlement is being mapped using drones, LiDAR (Light Detection and Ranging) technology and advanced digital surveying systems as part of the Adani Group-led Dharavi Redevelopment Project.

The result is a highly detailed three-dimensional model that captures lanes, rooftops, workshops, utility networks, access routes and public spaces. In effect, a digital version of the settlement is being preserved even as plans move forward to reshape its physical future.

Beyond Technology Alone

Unlike most global applications of digital twin technology, which are typically used in planned cities, airports and industrial facilities, Dharavi presents a uniquely complex challenge. It is a dense, organically evolved settlement where homes, workshops, businesses and community spaces have co-existed and adapted over several decades.

That complexity makes planning exceptionally difficult.